

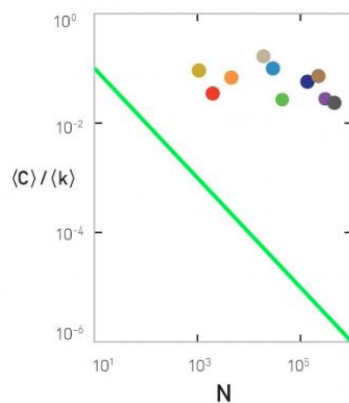
L5: Clustering in $G(n,p)$ networks

How large is the expected clustering coefficient at a random ER network (*i.e.*, a network constructed using the $G(n,p)$ model from Lesson-3)?

Recall that any two nodes in that model are connected with the same probability p . So, the probability that a connected triplet A-B-C forms a triangle (A-B-C-A) is also p . Thus, the expected value of the clustering coefficient for any node with more than one connection is p . Similarly, the transitivity (*and the average clustering coefficient*) of the whole network is also expected to be p .

In $G(n,p)$, the average degree is $\bar{k} = p(n-1)$. So, if the average degree remains constant, we expect p to drop inversely proportional with n as the network size grows. This means that if real networks follow the $G(n,p)$ model, we would see a decreasing trend between the clustering coefficient and the network size.

a. All Networks



This is not the case with real networks however. The plot shows the average clustering coefficient, normalized by the average degree, as a function of the network size N . In the $G(N,p)$ model, this normalized version of the average clustering coefficient should be equal to $1/N$ (shown as a green line in the plot). The various colored dots show the clustering coefficient for several real-world networks that appear in Table 3.2 of your textbook. These networks have widely different sizes, ranging from a thousand nodes to almost a million. Note that their clustering coefficient does not seem to get smaller as the network size increases.

The main message here is that the $G(n,p)$ model predicts negligible clustering, especially for large and sparse networks. On the contrary, real-world networks have a much higher clustering coefficient than $G(n,p)$ and its magnitude does not seem to depend on the network size.

Food For Thought

The transitivity (*and clustering coefficient*) focuses on a microscale property of a network, in the sense that it shows how likely it is that connected triplets form triangles. How would you explain that the value of this metric does not seem to depend on the size of the overall network in practice? What does that imply about the mechanisms that guide the formation of such networks?